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SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct

I YOGASHREE G (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Yashree

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data centers, and distributed servers.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

Expected concept nodes:

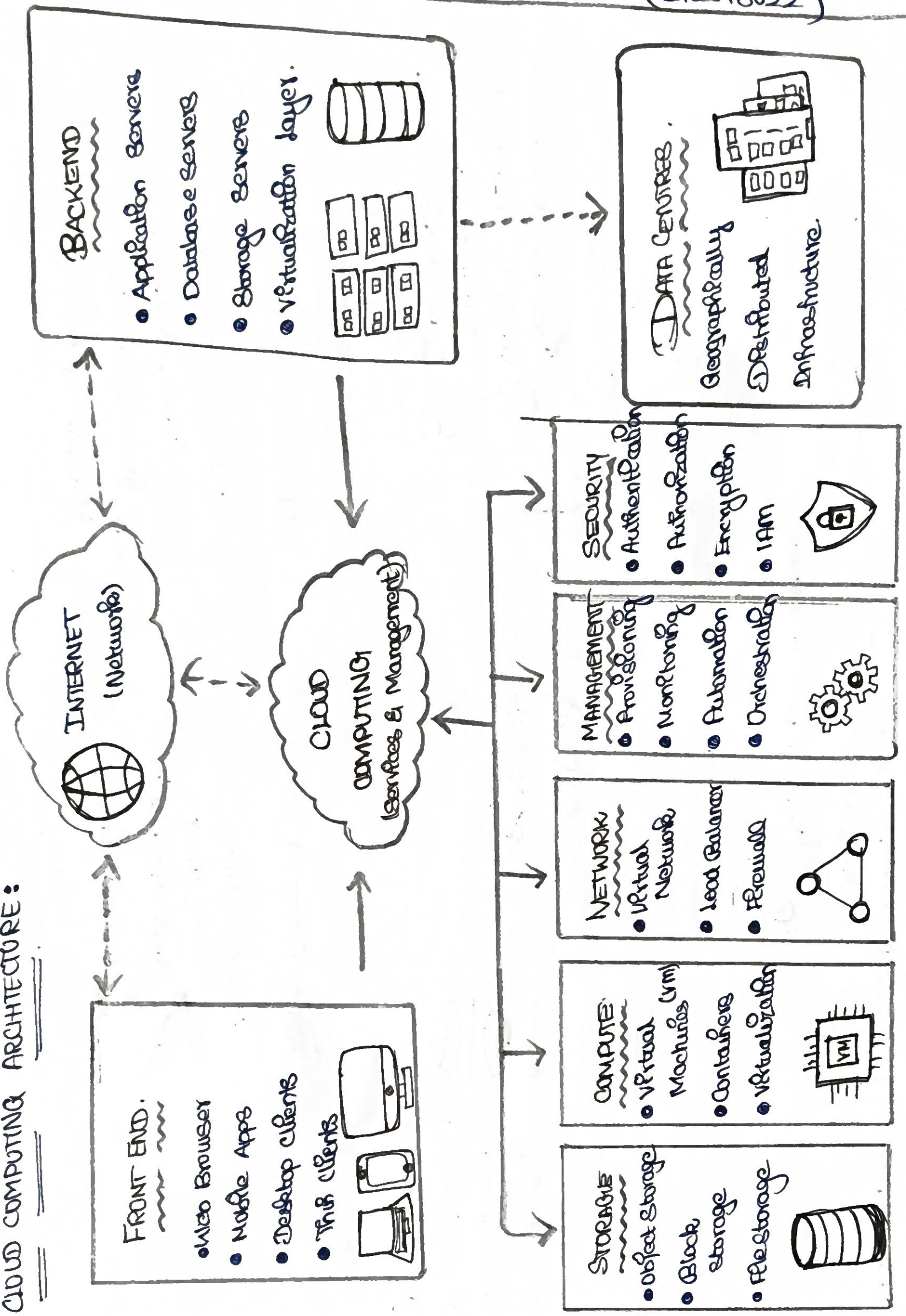
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

CLOUD COMPUTING ARCHITECTURE:

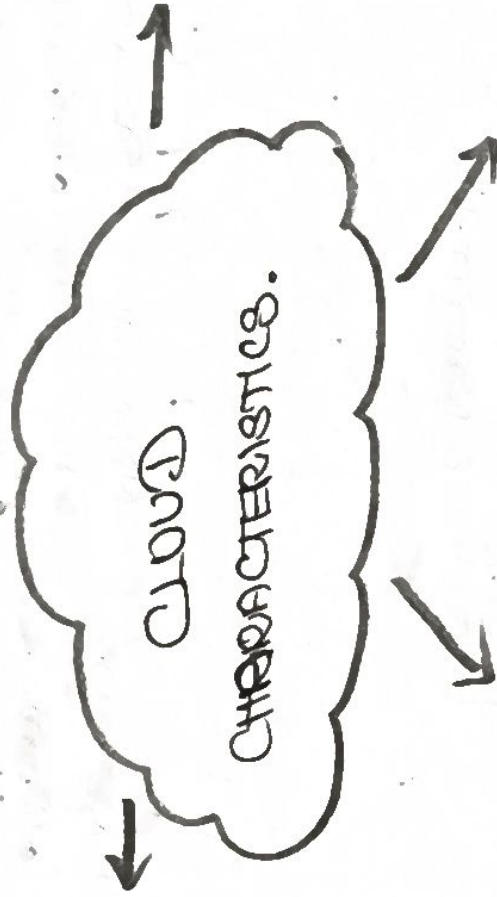
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CHARACTERISTICS OF CLOUD COMPUTING.

ON-DEMAND SELF-SERVICE
 Users can provision computing resources automatically without human interaction.

BROAD NETWORK ACCESS
 Services are available over the network and accessed through standard mechanisms.

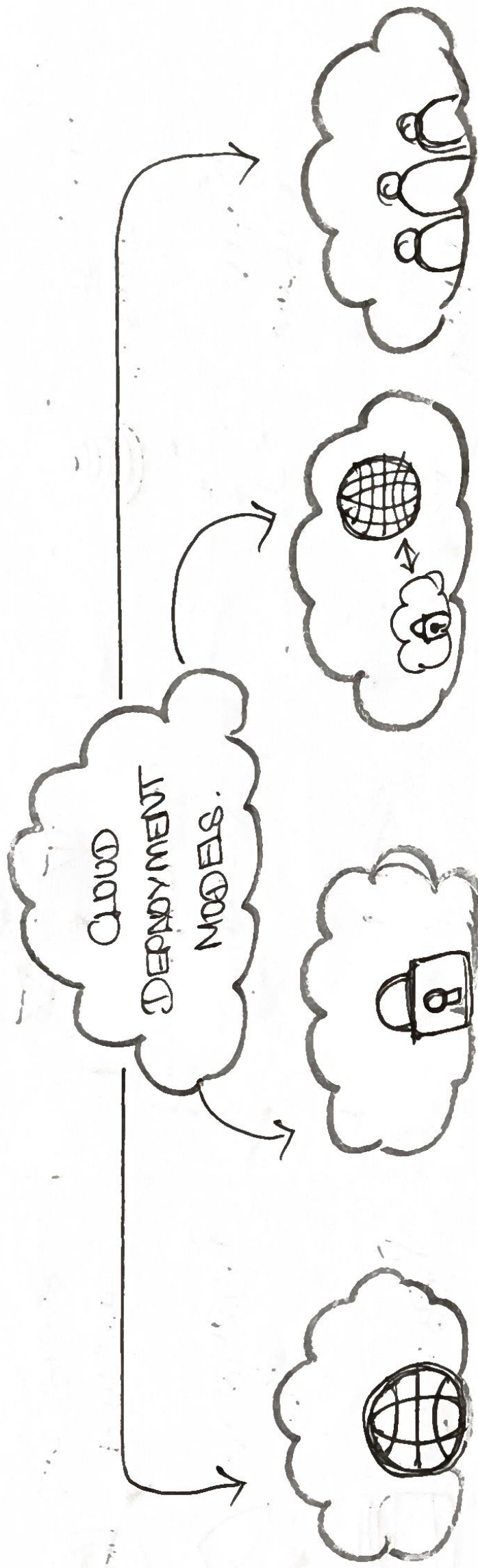



RESOURCE POOLING
 Provider's resources are pooled to serve multiple customers using a multi-tenant model.

RAPID ELASTICITY
 Resources can be scaled up or down quickly to match demand.


MEASURED SERVICE
 Resources usage is monitored, controlled, and reported for transparency and pay-per-use.


5) CLOUD DEPLOYMENT MODELS:



Public Cloud.

Services offered over the public internet by third party providers.

Examples: AWS, Microsoft Azure, Google Cloud.

Private Cloud.

cloud infrastructures used exclusively by a single organization.

Example: Openstack, VMware, Microsoft, Azure Stack.

Hybrid Cloud.

Combination of public and private clouds with data and application portability.

Example: AWS Outposts, Azure Hybrid, Google

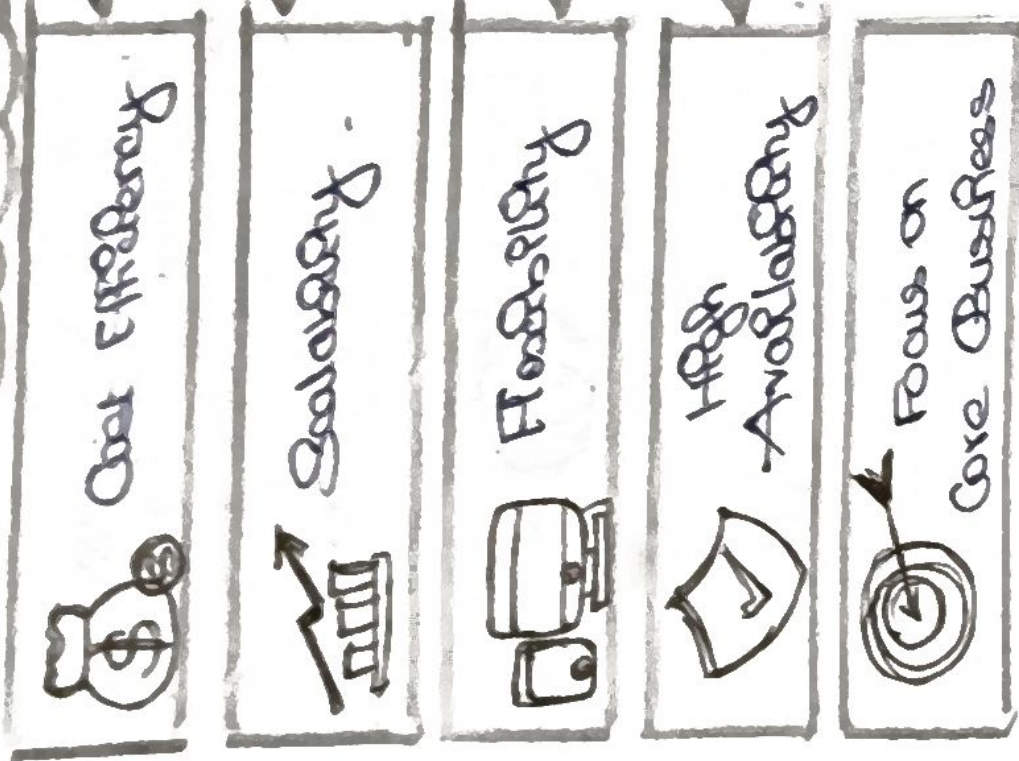
Community Cloud.

Shared by several organizations with common concerns (eg: Security, compliance.)

Example: Gov cloud, Healthcare Cloud.

4) CLOUD BENEFITS vs CHALLENGES

BENEFITS



CLOUD computing.

Balanced Approach



CHALLENGES

